Additional links and prototypes

Namma Space, Round 1 · Team Claudus (TM-60628EED289) · Techfest IIT Bombay 2026

Contents of this Drive folder

File	What it is
Technical_Abstract.pdf	System architecture, pipeline design, SOP construction and technical approach (5 pages or fewer)
Additional_Links_and_Prototypes.pdf	This document: every link in one place
Namma_Space_Demo.mp4	3 to 5 minute demonstration video: scope, capture process, 3D web walkthrough
Code_Repository/ (folder)	Code repository: `README.md` with setup instructions, `namma-space-code.zip` (scripts, viewer source, tests) and `namma-space-test-sample-data.zip` (the `test/sample_frames` data)

Proof of concept: live web viewer (free roaming)

https://namma-space-viewer-eosin.vercel.app

Open the link, wait for the model to load, press **Enter**. Walk with **W A S D** or the arrow keys, look with the mouse (click to capture it), hold **Shift** to sprint, press **C** for the calibration panel. On a phone, drag the left half of the screen to walk and the right half to look. No installation or account is needed. The viewer works in current Chrome, Edge and Firefox (WebGL2).

Code repository

Folder `Code_Repository` in this Drive folder. Start with `README.md`: it has the exact commands to reconstruct a model from a video (frame extraction, camera poses, splat training, clean-up, viewer). Unzip `namma-space-code.zip`, then unzip `namma-space-test-sample-data.zip` into the same folder to get the `test/` directory with 36 sample frames; `python test/run_test.py` registers them with COLMAP. The reconstructed model used by the live viewer is https://namma-space-viewer-eosin.vercel.app/assets/scene.spz (15 MB), with its placement file at /assets/config.json; both go into `viewer/assets/`.

Demonstration video

File `Namma_Space_Demo.mp4` in this folder.

Open-source components (also credited in the abstract)

Component	Licence	Link
COLMAP / pycolmap	BSD-style	github.com/colmap/colmap
Brush	Apache-2.0	github.com/ArthurBrussee/brush

Spark	MIT	github.com/sparkjsdev/spark
three.js	MIT	github.com/mrdoob/three.js
FFmpeg	LGPL/GPL	ffmpeg.org